

Application Form Vrije Universiteit Amsterdam

Application to a Master's Degree programme for International Degree Holders

Instructions:

- Please put all the required documents in this format.
- Read the instructions first, then delete the text and insert the requested documents.
- It is **not** possible to include external links.
- When you have added all documents, upload the form in **.PDF format** in the *Complete Registration* section of [your personal dashboard](#) on vu.nl.
- Only submit an application file that includes all required documents in this template.
- Please make sure to check the [relevant application deadline](#) to make sure you hand in your application on time.

Important: you need to be aware that this programme is fundamentally mathematical and is generally not suitable for students from a business or management background. You can check your eligibility for the master's programme in Business Analytics [here](#).

Details regarding your *previous or current* education

If you have completed multiple degrees at university level (Bachelor and Master), make sure to mention all of them and number them accordingly.

Country of your previous education	Romania
Name of the university	Alexandru Ioan Cuza University of Iași Faculty of Economics and Business Administration
Title and subject of the degree(s) in English <u>as well</u> as in the original language (for example (Honours) Bachelor of Science in Computer Science, Bachelor of Arts in History, Laurea Triennale in Economia, Grado en Estudios Internacionales etc.)	Bachelor of Economic Science in Business Information Systems
The (expected) date of graduation (needs to be by 31 August if you are planning to start in September)	July 2026
Nominal duration of your programme(s) / degree(s) in years (number of years from start to completion)	3 years
Total credits needed to complete your degree (no need to convert it to ECTS)	180 credits
Is a thesis part of your curriculum?	Yes

Your current or final GPA (Grade Point Average), also include the maximum possible GPA in your university's system (for example: 8/10, 3.27/4, etc.)	Currently, I finished the first semester of the third(final) year of my Bachelor, that's why I have access only to GPA from the first five semesters. So, my current GPA is: 8.33/10.
<u>English language test results</u> (If not obtained yet, please read the requirements carefully and indicate the expected date of submission)	July/August 2026

Curriculum Vitae (CV)

A curriculum vitae that includes your personal details and lists your previous education and relevant work experience. You can for example use [an Europass CV template](#).

Statement of Intent (Letter of Motivation)

A 1-2 page letter stating your reasons for applying to the programme (if applicable, please specify the **specialization** you are applying for).

You are required to indicate in your motivation letter how you meet the following admissions criteria:

- 90 ECTS or equivalent in Mathematics/Econometrics/Computer Science at university Bachelor's level, or equivalent of which:
 - 45 ECTS in Mathematics, including a Probability Theory course (6 ECTS) and a Statistics course (6 ECTS);
 - A minimum of 6 ECTS in Programming: Java, Python, C++ or another formal programming language that covers the OOP paradigm

If you apply with less than 45 ECTS in Mathematics, please explain in your motivation letter why you may still be eligible. In order to have a better insight into the level of Mathematics in this Master's programme, please see [these sample exams](#).

Please note: a Dutch 3-year Bachelor of Science contains a total of 180 ECTS (3 x 60 ECTS). This means that in order for you to meet the first requirement, you need to have at least approximately 1.5 years' worth of Mathematics/Econometrics/ Computer Science courses at the Bachelor's level.

English language test results

Optional document at this stage of the application.

You can already apply without having the English language test results and you do **not** need to include them in this application template.

If you already have **valid and sufficient** test results, please **upload** them via your VU personal dashboard in the separate category.

The **final deadline** for sending us the English Language test is **1 June (non-EEA) and 31 August (EEA)**. However, we **recommend** students to not wait until the final deadline and to make sure to hand in their English Language test no later than **1 May (non-EEA) and 1 August (EEA)**.

Please note that the minimum required scores can **differ** per Master's programme. For all relevant information on our English language test requirements, please visit [this page](#).

Diploma

Optional document at this stage of the application.

If you have already graduated, please insert a copy of your original Bachelor's diploma. Please include all relevant diplomas if you have multiple Bachelor's and/or Master's diplomas.

If you are conditionally admitted, the final deadline for sending us the **certified copy of your diploma** (by postal mail) or a certified digital copy (please find instructions [here](#)) before 31 August. We recommend students to hand in this document well in advance. Full details on handing in the certified copy of your diploma can be found on [this page](#).

You need to have graduated by 31 August if you wish to start the following September. If you have not graduated on this date, it is not possible to start in the following September.

Transcript of records

Please insert your (most up to date) **official transcript of records** from your Bachelor's and/or Master's programme here. If your university cannot provide you with a transcript in English, then please add your own translation as well.

Please include all your grades; this means all semesters/years and from every Bachelor's and/or Master's programme you have completed.

In case you went on **exchange** during your programme and the courses and grades are not mentioned on your main transcript, please add the transcript from your exchange university as well.

Please make sure that your transcript mentions a current grade point average (GPA). Please also include a grading system explanation. If you do not provide a GPA, if needed we will calculate your GPA on the information provided.

Course description including used literature

Details regarding your previous or current education	
<i>Please do not submit your programme's official syllabus/curriculum overview.</i>	
Name programme(s)/degree(s)	Bachelor of Economic Science in Business Information Systems
Nominal duration of your programme(s) / degree(s) in years	3 years
Total credits needed to complete your degree (no need to convert it to ECTS)	180 credits
Total credits in mathematics	25 credits
Total credits in econometrics	5 credits
Total credits in computer science	61 credits

Please describe the courses you have taken during your previous higher education that are relevant to the Master's programme you are applying for.

The **relevant courses for admission** can be found on the [programme pages](#), under the **tab 'Admissions'**, section '**I have an international degree**'.

Also clearly mention which relevant courses you will still complete before graduation.

Finally, partition the courses over the key fields of:

- Mathematics
- Econometrics
- Computer Science

If you need more space to include all relevant courses, you can copy paste the format below.

Relevant courses for intended Master's programme in the field of:

Mathematics (incl Probability & Statistics)

Course name	Applied Mathematics in Economy	
Bachelor year (1, 2, 3, still to be completed etc.)	1	
Credits or study hours	<u> 5 </u> credits	<u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Quantitative Foundations: Acquired a rigorous mathematical framework using linear algebra concepts (vector spaces, matrix operations, quadratic forms) essential for handling multidimensional datasets • Optimization Mastery: Mastered Linear Programming and the Simplex algorithm to solve resource allocation problems and maximize objective functions, a core skill for prescriptive analytics • Logistics & Operations: Applied algorithms for Transportation Problems (e.g., Least Cost Method, Stepping-stone) to optimize supply chain efficiency and minimize costs • Calculus for Analysis: Utilized the Hessian matrix and partial derivatives to analyze marginal changes and determine local extrema for profit maximization strategies • Statistical Modeling: Implemented the Least Squares Method to perform linear regression and fit mathematical models to economic data, providing a basis for predictive analysis • Modeling Logic: Developed the critical skill of translating business scenarios into mathematical models, identifying constraints and objectives for data-driven decision-making	

Used literature / textbook	• Sydsæter, K., Hammond, P., “Essential Mathematics for Economic Analysis” – 3rd edition, Prentice-Hall, Inc., U.K., 2008 • Sydsæter, K., Hammond, P., et al., “Further Mathematics for Economic Analysis” – 2nd edition, Prentice-Hall, Inc., U.K., 2008 • Anton, H., “Elementary Linear Algebra”, 5th edition, WIE, New York, 1987 • Diaconița, V., Rusu, Gh., Spînu, T.M., “Matematici aplicate în economie”, Ed. Sedcom Libris, Iași, 2004
Course name	Operations Research
Bachelor year (1, 2, 3, still to be completed etc.),	2
Credits or study hours	<u> 5 </u> credits <u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Constrained Optimization: Applied multivariate calculus and the Method of Lagrange Multipliers to solve complex economic optimization problems involving multiple constraints, determining local and global extrema • Stochastic Modeling: Modeled dynamic economic systems using Markov Chains (finite and absorbing states) to analyze transition probabilities and predict long-term equilibrium states in market behaviors • Network Optimization: Mastered Graph Theory algorithms, including the Ford-Fulkerson algorithm for maximum flow problems and Kruskal’s algorithm for minimum spanning trees, essential for logistics and supply chain analytics • Project Management Analytics: Implemented the Critical Path Method (CPM) and optimal path algorithms (Bellman-Kalaba, Dantzig) to optimize project scheduling and resource routing • Algorithmic Decision Making: Developed the competency to translate operational business problems into mathematical models (graphs,

	trees, stochastic processes) and solve them using rigorous algorithmic logic	
Used literature / textbook	• Goldstein, L.J., Schneider, D.I., Siegel, M.J., "Finite mathematics & its applications", 10th edition, Pearson Prentice Hall, USA, 2010 • Barnett, R. A., Ziegler, M. R., et al., "Finite Mathematics for Business, Economics, Life Sciences and Social Sciences", 11th edition, Prentice-Hall, Inc., USA, 2008 • Sydsæter, K., Hammond, P., "Essential mathematics for economic analysis", 3rd edition, Pearson, FT Prentice Hall, 2008 • Spînu, T.M., "Cercetări operaționale", Course Notes, Portal FEAA, Iași, 2015	
Course name	Basis of Statistics	
Bachelor year (1, 2, 3, still to be completed etc.),	1	
Credits or study hours	<u> 6 </u> credits	<u> 150 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Probability Theory Foundations: Mastered the theoretical framework of probability distributions, distinguishing between discrete models (Bernoulli, Binomial) and continuous models, which are fundamental for risk analysis in business • Theoretical Distributions: Gained in-depth knowledge of key statistical distributions used in analytics, specifically the Normal (Gaussian) distribution, Chi-square (χ^2), Student-t, and Fisher, understanding their properties and applications in modeling economic phenomena • Statistical Inference Logic: Applied probability theory to perform parameter estimation,	

	calculating point estimates and confidence intervals for population means, proportions, and variances • Hypothesis Testing: Utilized theoretical distributions to conduct rigorous hypothesis testing, covering single-sample tests, two-sample comparisons, and analysis of variance (ANOVA) for multiple independent samples • Data Analysis & Exploration: Performed univariate and bivariate analysis to describe data distribution (central tendency, dispersion, shape) and analyze relationships between variables using correlation and regression logic	
Used literature / textbook	• Glenberg, A.M., Andrzejewski, M.E., "Learning from data. An Introduction to Statistical Reasoning", Taylor and Francis Group, New York, 2008 • Wonnacott, H., Wonnacott, T., "Statistiques" (Statistics), Economica, Paris, 1991 • Pintilescu, C., "Analiza statistică a datelor în SPSS și în R", Editura Sedcom Libris, Iași, 2022 • Jaba, E., "Statistica", 3rd edition, Ed. Economică, București, 2002	
Course name	Financial-Accounting Management of the Enterprise	
Bachelor year (1, 2, 3, still to be completed etc.),	3	
Credits or study hours	__4__ credits	__100__ study hours
Course description or learning outcomes (5-10 sentences)	• Deterministic Inventory Optimization: Applied mathematical models to determine the optimal level of stocks within the operating cycle, focusing on cost minimization: • Capital Structure Optimization: Utilized quantitative criteria to identify and configure the optimal financial structure and leverage for an organization	

	• Investment Efficiency Analysis: Applied financial-mathematical formulas to assess the efficiency and profitability metrics of long-term investment projects • Quantitative Financial Forecasting: Mastered analytical methods for forecasting enterprise financing needs and managing cash flow through data-driven planning
Used literature / textbook	• Ross, S. A., Westerfield R. W., Jaffe J. F., "Corporate Finance", Dunod Paris, 2005 • Berk, J., DeMarzo, P., "Corporate Finance", Pearson Education, Paris, 2008 • Bucataru, D., "Gestiunea financiară a întreprinderii" (Corporate Financial Management), 3rd edition, Junimea Publishing, Iași, 2006 • Mironiuc, M., "Scientific Foundations of Corporate Financial-Accounting Management", 3rd edition, Alexandru Ioan Cuza University Press, Iași, 2018

Relevant courses for intended Master's programme in the field of:

Econometrics

Course name	Econometrics	
Bachelor year (1, 2, 3, still to be completed etc.), Master year	2	
Credits or study hours	<u>5</u> credits	<u>125</u> study hours
Course description or learning outcomes (5-10 sentences)	• Regression Modeling: Mastered the Ordinary Least Squares (OLS) method to estimate and interpret Simple and Multiple Linear Regression models, quantifying relationships between economic variables • Model Validation: Learned to validate statistical models by rigorously testing the assumptions of the Classical Linear Regression Model (Gauss-Markov), identifying issues like multicollinearity or heteroscedasticity • Statistical Inference: Conducted advanced hypothesis testing (t-tests, F-tests) to assess the statistical significance of regression coefficients and the overall explanatory power of the models • Non-linear Analysis: Addressed complex data patterns by estimating and interpreting non-linear models, expanding the analytical toolkit beyond strict linearity • Applied Analytics: Applied econometric techniques to real-world datasets using statistical software (SPSS/R), translating raw data into actionable insights for decision-making • Predictive Logic: Developed the ability to use estimated models for prediction and forecasting, calculating point estimates and confidence intervals for specific business scenarios	

Used literature / textbook	• Gujarati, D.N., "Basic Econometrics", 4th Edition, McGraw-Hill, 2004 • Greene, W.H., "Econometric Analysis", 4th Edition, Prentice Hall, 2000 • Stock, J.H., Watson, M.W., "Introduction to Econometrics", Pearson International Edition, 2007 • Jemna, D., "Econometrie cu aplicatii in R" (Econometrics with applications in R), Editura Universităţii „Al.I. Cuza”, Iaşi, 2017	
Relevant courses for intended Master's programme in the field of:		
Computer Science (including a programming course)		
Course name	Algorithms and Programming Logic	
Bachelor year (1, 2, 3, still to be completed etc.),	2	
Credits or study hours	__6__ credits	__150__ study hours
Course description or learning outcomes (5-10 sentences)	• Algorithmic Efficiency: Mastered the analysis of algorithmic complexity (Big O notation) to evaluate time and space trade-offs, a critical skill for processing large-scale business datasets efficiently • Data Structures: Implemented and manipulated advanced data structures including Hash Tables (Maps), Linked Lists, Stacks, and Queues to optimize data storage and retrieval speeds • Optimization Strategies: Applied the Greedy Method (e.g., Dijkstra's algorithm) and Backtracking to solve optimization problems such as shortest path finding and resource allocation • Network Analysis: Studied Graph Theory concepts and algorithms (BFS, DFS, Minimum Spanning Trees) which are foundational for analyzing network relationships, logistics, and supply chains	

	• Pattern Matching: Gained proficiency in String processing and Regular Expressions (RegEx), essential techniques for text mining, data cleaning, and unstructured data analysis • Practical Implementation in Java: Coded all studied algorithms and data structures from scratch using Java, utilizing Object-Oriented Programming principles (interfaces, inheritance, generics) to build robust, reusable, and efficient software solutions	
Used literature / textbook	• Cormen, T., Leiserson, C., Rivest, R., Stein, C., "Introduction to Algorithms", 3rd Edition, MIT Press Cambridge, England, 2009 • Sedgewick, R., Wayne, K., "Algorithms", 4th Edition, Princeton University, Addison-Wesley, 2011 • Goodrich, M.T., Tamassia, R., "Data Structures and Algorithms in Java", 6th Edition, Wiley, 2014 • Erickson, J., "Algorithms", University of Illinois, 2015	
Course name	Programming I	
Bachelor year (1, 2, 3, still to be completed etc.),	2	
Credits or study hours	<u> 6 </u> credits	<u> 150 </u> study hours
Course description or learning outcomes (5-10 sentences)	• C# Development Foundations: Developed strong programming skills using C# within the .NET framework and Visual Studio, mastering data types, variables, and control structures for robust software development • Data Structures Management: Implemented and manipulated complex data collections including vectors, multi-dimensional arrays, lists, queues, and stacks to organize and process business data efficiently • Database Integration: Built data-driven applications by establishing connections to databases and performing CRUD operations	

	(Insert, Delete, Update, Read), bridging the gap between software logic and data storage • Object-Oriented Design: Applied Object-Oriented Programming (OOP) principles to design scalable application components, ensuring code reusability and modularity • Reporting & GUI: Designed Graphical User Interfaces (GUI) using Windows Forms and developed automated reports, translating raw data into user-friendly visual insights • Algorithmic Logic: Implemented searching (Binary Search) and sorting algorithms directly in C#, optimizing the performance of data retrieval processes
Used literature / textbook	• Michaelis, M., "Essential C# 12.0 (Addison-Wesley Microsoft Technology Series)", 8th Edition, Addison-Wesley Professional, 2023 • Farrell, J., "Microsoft Visual C#: An Introduction to Object-Oriented Programming", 8th Edition, Cengage Learning, 2024 • Dospinescu, O., "Programarea în C#. Structuri, interfețe, baze de date, rapoarte", Editura Tehnopress, 2013
Course name	Programming II
Bachelor year (1, 2, 3, still to be completed etc.),	3
Credits or study hours	<u> 5 </u> credits <u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Java Ecosystem Mastery: Developed advanced software solutions using the Java language and platform, mastering Object-Oriented Programming (OOP) principles, polymorphism, and generic programming for scalable applications • Data Processing Pipelines: Implemented Functional Programming concepts using Lambda expressions and the Java Streams API to create

	efficient data processing pipelines (filter, map, reduce, grouping) for complex datasets • Advanced Data Structures: Utilized the Java Collections Framework (Lists, Sets, Maps/HashMaps) to store, manipulate, and retrieve data efficiently, optimizing algorithmic performance • Database Connectivity (SQL & ORM): Built robust data persistence layers using JDBC and JPA (Java Persistence API) to connect Java applications with PostgreSQL databases, performing advanced CRUD operations and object-relational mapping • Design Patterns: Applied industry-standard Design Patterns (Singleton, Factory, Builder) to solve recurring architectural problems and ensure code maintainability • Application Architecture: Designed layered applications (MVC architecture) for both Desktop (JavaFX) and Web environments, separating business logic from data presentation
Used literature / textbook	• Bloch, J., "Effective Java", 3rd Edition, Addison-Wesley, 2018 • Schildt, H., "Java: A Beginner's Guide", 9th Edition, McGraw Hill, 2022 • Sierra, K., Bates, B., Gee, T., "Head First Java", 3rd Edition, O'Reilly Media, 2022 • Mihalcea, V., "High-Performance Java Persistence", Lean Publishing, 2020

Course name	Databases	
Bachelor year (1, 2, 3, still to be completed etc.),	2	
Credits or study hours	<u>6</u> credits	<u>125</u> study hours
Course description or learning outcomes (5-10 sentences)	• Relational Database Design: Learned the principles of database architecture, applying Normalization techniques (1NF to 5NF) to eliminate redundancy and ensure data integrity in business schemas • Advanced SQL Querying in PostgreSQL: Mastered SQL syntax within the PostgreSQL environment, writing complex queries that utilize multi-table joins (inner/outer), subqueries, and set operators for data retrieval • Data Aggregation & OLAP: Implemented advanced data summarization techniques using aggregate functions, grouping (GROUP BY), and OLAP (Online Analytical Processing) options for multidimensional business reporting • R & Tidyverse Integration: Integrated SQL databases with R (via RStudio), performing seamless data import/export and utilizing the tidyverse ecosystem for further data cleaning and manipulation • Data Manipulation (DML): Executed data manipulation operations (INSERT, UPDATE, DELETE) and managed transactions to ensure the consistency of large datasets • Analytical Logic: Developed the ability to translate business questions into optimized SQL queries and R scripts, bridging the gap between raw data storage and analytical insights	
Used literature / textbook	• Wickham, H., Grolemund, G., "R for Data Science", 2nd edition, O'Reilly, 2022 • Fotache, M., "SQL. Dialecte DB2, Oracle, PostgreSQL și SQL Server", Ed. Polirom, Iași, 2009 • Fotache, M., "Data Processing, Analysis and Science with R", GitHub Resources, 2023	

	• Fotache, M., "Proiectarea bazelor de date. Normalizare și postnormalizare", Ed. Polirom, Iași, 2005	
Course name	Information Technologies for Business	
Bachelor year (1, 2, 3, still to be completed etc.),	1	
Credits or study hours	<u> 5 </u> credits	<u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Digital Business Context: Analyzed the profound impact of the digital world on the modern organizational environment, understanding how IT infrastructure drives business strategy and operational efficiency • Information Systems Architecture: Gained a solid theoretical understanding of Information Systems (IS) within organizations, including an introduction to Enterprise Resource Planning (ERP) concepts and their role in data integration • Business Productivity Tools: Mastered advanced features of office productivity software (Microsoft Office suite) to solve specific business problems, manage documentation, and deliver professional graphical presentations • Web Technologies in Business: Explored the application of Internet technologies and the World Wide Web in business administration, understanding the infrastructure that supports e-commerce, cloud computing, and digital communication • IT Ethics & Security: Addressed the social and ethical aspects of Information and Communication Technologies (ICT), fostering a responsible approach to data privacy, security protocols, and intellectual property in a business context	
Used literature / textbook	• O'Brien, J., "Introduction to Information Systems", McGraw-Hill Irwin, Boston, MA, 2003	

	• Airinei, D., Grama, A., Fotache, D., et al., "Tehnologii informaționale aplicate in organizatii", Editura Universității „Alexandru Ioan Cuza”, Iași, 2014 • Grama, A. (coord.), "Tehnologii informaționale în lumea afacerilor", Editura Sedcom Libris, Iași, 2009	
Course name	Business Software Tools	
Bachelor year (1, 2, 3, still to be completed etc.),	1	
Credits or study hours	<u>6</u> credits	<u>124</u> study hours
Course description or learning outcomes (5-10 sentences)	• Data Structures & Algorithms: Gained a foundational understanding of data structures and processing algorithms, essential for organizing business information logic before implementation • Relational Database Design: Mastered the theoretical concepts of the Relational Model, learning how to structure data efficiently into connected tables to minimize redundancy • Database Management Systems (DBMS): Utilized Microsoft Access to create and manage databases, defining relationships between entities and enforcing data integrity rules • SQL Querying: Acquired proficiency in Structured Query Language (SQL) to execute queries for data extraction, filtering, and manipulation directly from the database, a core skill for data analytics • Advanced Spreadsheet Modeling: Applied advanced Microsoft Excel functions to process economic data, perform simulations of business transactions, and visualize results through reporting	
Used literature / textbook	• Airinei, D., Filip, M., Grama, A., Fotache, M., et al., "Instrumente software pentru afaceri"	

	(Business Software Tools), Editura Sedcom Libris, Iași, 2007 • Airinei, D., Dospinescu, O., et al., "Instrumente software pentru afaceri. Aplicații practice" (Business Software Tools. Practical Applications), Editura Sedcom Libris, Iași, 2010 • Airinei, D., Filip, M., et al., "Instrumente software pentru afaceri. Teste grilă" (Business Software Tools. Quizzes), Editura Sedcom Libris, Iași, 2007	
Course name	Office Information Systems	
Bachelor year (1, 2, 3, still to be completed etc.),	1	
Credits or study hours	<u> 4 </u> credits	<u> 100 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Advanced Data Analysis in Excel: Mastered advanced functions in Microsoft Excel, specifically using Pivot Tables for data consolidation and Solver for optimization scenarios, essential for solving linear programming problems in business • Project Management Skills: Acquired practical skills in project planning using Microsoft Project (and Project Libre), learning to schedule tasks, manage critical paths, and allocate resources efficiently • Digital Data Collection: Implemented data gathering strategies using Google Forms and Google Drive, understanding how to design questionnaires for market research and manage cloud-based collaborative workflows • Document Automation: Utilized advanced document processing features in Microsoft Word (Mail Merge, electronic forms, templates) to streamline administrative workflows and generate mass business communications • Information Systems Theory: Gained a solid understanding of Office Information Systems	

	(OIS) and their role in the digital economy, including concepts of E-commerce, Electronic Data Interchange (EDI), and digital payment systems • IT Security & Ethics: Explored the fundamental principles of protecting digital assets and personal data (GDPR context), understanding the security risks associated with modern office environments	
Used literature / textbook	• Piccoli, G., "Information Systems for Managers: Texts and Cases", 3rd Edition, Wiley, 2016 • Weisbord, M., "Productive Workplaces", Jossey Bass Wiley, 2012 • Marmel, E., "Teach Yourself VISUALLY Office 2013", Visual 1 Edition, 2013 • Georgescu, M., Popescu, D., "Sisteme informaționale de birou pentru administrația publică", Iași, 2013	
Course name	Local Computer Networks	
Bachelor year (1, 2, 3, still to be completed etc.),	3	
Credits or study hours	<u> 5 </u> credits	<u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Network Architecture: Mastered the layered architecture of networks (OSI and TCP/IP models), analyzing how data packets traverse local and global networks to support distributed business applications • Data Security & Access: Studied network security principles and implemented policies for user authentication (LDAP) and access control, ensuring the confidentiality and integrity of organizational data • Virtualization Technologies: Acquired practical skills in virtualization, installing and managing Virtual Machines (VMs), which serves as the	

	foundational knowledge for modern Cloud Computing environments • Network Administration: Performed practical network configuration and troubleshooting using command-line tools (ping, ipconfig) and managed server roles to optimize resource sharing within an organization • Infrastructure Design: Learned the physical and logical design of Local Area Networks (LANs), including the selection of transmission media (fiber optics, wireless) and budgeting for IT infrastructure acquisitions	
Used literature / textbook	• Kurose, J.F., Ross, K.W., "Computer Networking: A Top-Down Approach", 7th Edition, Pearson, 2017 • Tanenbaum, A.S., Wetherall, D.J., "Computer Networks", 5th Edition, Pearson, 2011 • Munteanu, A., Greavu-Şerban, V., "Reţele locale de calculatoare", Editura Universităţii „Alexandru Ioan Cuza”, Iaşi	
Course name	E-Commerce	
Bachelor year (1, 2, 3, still to be completed etc.),	3	
Credits or study hours	<u>5</u> credits	<u>180</u> study hours
Course description or learning outcomes (5-10 sentences)	Web Development Fundamentals: Acquired practical skills in front-end web development by writing code in HTML and CSS to structure and style web pages, understanding the technical anatomy of an online store • E-Commerce Platforms (Shopify): Gained hands-on experience in building and managing e-commerce sites using Shopify, configuring product catalogs, payment gateways, and analyzing back-end store logic	

	• Scripting & Interactivity: Introduction to JavaScript and jQuery to add dynamic features to web interfaces, enhancing user experience (UX) and site functionality • Business Models & Strategy: Analyzed various e-commerce business models (B2B, B2C) and developed digital strategies for market entry, branding, and customer retention • Digital Marketing & Conversion: Studied digital marketing techniques and conversion optimization strategies essential for driving traffic and sales in a competitive online environment • Security & Integration: Understood the critical aspects of e-commerce security (payment protection) and the integration of online stores with enterprise systems like ERP and CRM for data synchronization	
Used literature / textbook	• Brynjolfsson, E., Urban, G., "Strategies for E-Business Success", Wiley, San Francisco, 2001 • Carey, P., "New Perspectives on HTML, XHTML, and XML", 3rd Edition, Cengage Learning, 2009 • Haig, M., "The B2B E-commerce Handbook", Kogan Page, London, 2001 • Matthewson, J., "e-Business - A Jargon-Free Practical Guide", Routledge, Oxford, 2012	
Course name	Integrated Systems – ERP	
Bachelor year (1, 2, 3, still to be completed etc.),	Still to be completed	
Credits or study hours	__4__ credits	__100__ study hours
Course description or learning outcomes (5-10 sentences)	ERP Architecture & Data Integration: I will analyze how Enterprise Resource Planning (ERP) systems integrate core business processes (finance, production, HR) into a unified	

	architecture, understanding the flow of data required for enterprise-level analytics • Business Intelligence (BI): I will explore the next generation of decision-support systems by studying Business Intelligence modules within ERPs, learning how to extract actionable insights from raw operational data • Reporting & Data Visualization: I will gain practical experience with the WinMentor platform, specifically focusing on generating complex financial, commercial, and production reports that serve as the foundation for business performance analysis • Supply Chain & CRM Analytics: I will study the functional extension of ERPs into Supply Chain Management (SCM) and Customer Relationship Management (CRM), learning to analyze data across the entire value chain • ERP Platform Hands-on Practice: I will configure and operate Microsoft Dynamics AX (Axapta), covering accounting, financial management, HR, production, and distribution modules — gaining direct experience with an enterprise-grade ERP system used internationally • Project Management Implementation: I will learn the methodologies for managing ERP implementation projects, identifying risks and opportunities in organizational data restructuring
Used literature / textbook	• Bradford, M., "Modern ERP: Select, Implement, and Use Today's Advanced Business Systems", 4th Edition, USA, 2023 • Shibly, H.R., et al., "ERP Adoption in Organizations", Palgrave Macmillan, 2022 • Langenwalter, G., "Enterprise Resources Planning and Beyond", CRC Press, 2020

	• Hurbean, L., Fotache D., et al., "Platforme integrate pentru afaceri. ERP", Editura Economica, Bucuresti, 2013	
Course name	Information Systems Design	
Bachelor year (1, 2, 3, still to be completed etc.),	Still to be completed	
Credits or study hours	<u> 6 </u> credits	<u> 150 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Java Implementation Strategy: I will execute the technical implementation of the designed system using C#, translating logical specifications into functional object-oriented code to solve specific business problems • Logical Data Modeling: I will apply advanced conceptual modeling techniques to design the logical architecture of databases, ensuring data integrity and efficient storage for high-volume processing • System Development Life Cycle (SDLC): I will master the stages of the software development lifecycle, learning to define specifications and requirements for complex business information systems • Software Architecture: I will design the architecture of software components, understanding how to integrate databases, user interfaces, and processing algorithms into a cohesive solution • User Interface (UI) & Dashboard Design: I will learn principles of User Interface design to create intuitive forms and effective business reports, focusing on usability and data presentation for decision-makers • Team Project Management: I will collaborate in a team to develop a complete software component, simulating a real-world IT project environment and managing roles and responsibilities	

Used literature / textbook	• Mandel, T., "The Elements of User Interface Design", John Wiley & Sons, New York, 1997 • Kendall, K.E., Kendall, J.E., "Systems Analysis and Design", 5th Edition, Prentice Hall, 2002 • Pressman, R.S., "Software Engineering: A Practitioner's Approach", McGraw-Hill, 2000 • Teorey, T.J., "Database Modeling & Design", Morgan Kaufmann Publishers, Inc, 1999	
Course name	Information Systems Security	
Bachelor year (1, 2, 3, still to be completed etc.),	Still to be completed	
Credits or study hours	__5__ credits	__125__ study hours
Course description or learning outcomes (5-10 sentences)	• Risk Management Strategy: I will master the concepts of information risk management, learning to identify vulnerabilities and implement strategic policies to protect organizational data assets according to international standards like ISO 27001 • IT Governance & Compliance: I will study IT governance frameworks (specifically COBIT 5) and legal regulations to ensure data privacy and ethical handling of sensitive business information • Vulnerability Assessment: I will perform practical security assessments using simulation platforms (Cyber Range) and "Capture The Flag" (CTF) exercises to understand the anatomy of cyberattacks and how to mitigate them • Network Security Architecture: I will analyze network security protocols and layered defense strategies to prevent unauthorized access and data breaches in distributed business environments • Identity & Access Management (IAM): I will explore mechanisms for user authentication and authorization, ensuring that critical business	

	analytics data is accessible only to authorized personnel • Business Continuity Logic: I will develop the ability to design security policies that ensure business continuity and minimize the economic impact of security incidents	
Used literature / textbook	• ISACA, "CISM Review Manual", 16th Edition, ISACA, 2020 • ISACA, "COBIT 5 for Information Security", ISACA, 2012 • Bigdoli, H., "Handbook of Information Security", Vol. 1-3, John Wiley & Sons, 2006 • LeVeque, V., "Information Security – a Strategic Approach", Wiley Interscience, 2006 • Oprea, D., "Protectia si securitatea sistemelor informatinale", Editura Polirom, Iasi, 2007	
Course name	Specialized Internship	
Bachelor year (1, 2, 3, still to be completed etc.),	2	
Credits or study hours	<u> 4 </u> credits	<u> 120 </u> study hours
Course description or learning outcomes (5-10 sentences)	• IT & Digital Services Context: Completed a practical internship within an IT and Digital Services company, gaining hands-on experience in the operational workflows of software delivery and digital product management • Quality Assurance (QA): Performed manual testing on digital products, identifying functional bugs and reporting inconsistencies to ensure high-quality software releases • Jira & Agile Management: Utilized Atlassian Jira to manage the product backlog, track software defects, and monitor the resolution status of issues within an Agile development framework • User Stories & Requirements: Collaborated with the product team to define and refine User	

	Stories, translating client business needs into clear technical acceptance criteria for developers • Automated Testing: Designed and executed structured test cases using STLC methodology, ensuring systematic validation of software functionality and quality standards • Team Collaboration: Worked in a cross-functional team environment, participating in project phases from requirement gathering to final validation	
Used literature / textbook	• Atlassian, "Jira Software Documentation & User Guide", Official Online Documentation, 2023 • Cohn, M., "User Stories Applied: For Agile Software Development", Addison-Wesley Professional, 2004 • ISTQB, "Certified Tester Foundation Level Syllabus", International Software Testing Qualifications Board, 2018 • Airinei, D., Grama, A., et al., "Tehnologii informaționale aplicate in organizatii", Editura Universitatii "Alexandru Ioan Cuza" Iasi, 2014	
Course name	Web Technologies	
Bachelor year (1, 2, 3, still to be completed etc.),	Still to be completed	
Credits or study hours	<u> 5 </u> credits	<u> 125 </u> study hours
Course description or learning outcomes (5-10 sentences)	• Web Application Development: Building functional web applications using HTML5, CSS3, and JavaScript, including semantic markup, responsive design with Tailwind CSS/Media Queries, and DOM manipulation for dynamic user interfaces. • JavaScript Programming: Mastering core programming concepts in JavaScript — data types, functions, control structures, data structures, and object-oriented programming —	

	applied to solving business problems through code. • ReactJS Framework: Developing component-based web interfaces using ReactJS, including integration with external services and Node.js for server-side data processing. • Version Control & Dev Tools: Utilizing Git for source control and Visual Studio Code as development environment, applying industry-standard workflows for collaborative software development.
Used literature / textbook	• Musciano, Zeldman J., <i>"Designing with Web Standards"</i>, New Riders Press, 2005 • Kennedy B., <i>"HTML & XHTML: The Definitive Guide"</i>, 6th Edition, O'Reilly, 2006 • Meyer E., <i>"CSS: The Definitive Guide"</i>, 3rd Edition, O'Reilly, 2006 • Paul-Koch P., <i>"PPK on JavaScript"</i>, New Riders Press, 2006

Thesis (or another sample of academic writing)

Please insert your **thesis** here or, if you did not write a thesis (yet), an **alternative writing sample** such as an academic paper or essay (written in English). If you have not written anything in English during your studies, then please select one of your own academic writing samples and add a 2-5 page summary in English **before** the text in the original language. Always include your **literature references**. Please be aware that for some programmes you need to upload a specific essay as mentioned on that programme's webpage.

Please note we will always check application documents for plagiarism.